

IRAK4 - Target Landscape Report

Target Protein: IRAK4

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1. Target Profile

Protein Overview

IRAK4 is a serine/threonine-protein kinase with an accession number Q9NWZ3. It plays a critical role in initiating the innate immune response against foreign pathogens and is involved in Toll-like receptor (TLR) and IL-1R signaling pathways.

Molecular Function & Mechanism

IRAK4 functions by being rapidly recruited by MYD88 to the receptor-signaling complex upon TLR activation, forming the Myddosome together with IRAK2. It phosphorylates initially IRAK1, stimulating its kinase activity and promoting intensive autophosphorylation of IRAK1. IRAK4 also phosphorylates E3 ubiquitin ligases Pellino proteins (PELI1, PELI2, and PELI3), leading to their polyubiquitination of IRAK1. The ubiquitin-binding domain of IKBKG/NEMO binds to polyubiquitinated IRAK1, bringing together the IRAK1-MAP3K7/TAK1-TRAF6 complex with the NEMO-IKKA-IKKB complex, which activates IKKs (CHUK/IKKA and IKBKB/IKKB), leading to NF-kappa-B nuclear translocation and activation. Additionally, IRAK4 phosphorylates TIRAP for its ubiquitination and degradation, regulates NADPH oxidase activation after LPS stimulation, and interacts with NCF1.

Disease Associations

IRAK4 is implicated in Immunodeficiency 67, an autosomal recessive primary immunodeficiency characterized by recurrent, life-threatening systemic and invasive bacterial infections beginning in infancy or early childhood.

Signalling Pathways

IRAK4 participates in the MAPK signaling pathway, NF-kappa B signaling pathway, Toll-like receptor signaling pathway, NOD-like receptor signaling pathway, Neurotrophin signaling pathway, Alcoholic liver disease, Pathogenic Escherichia coli infection, Salmonella infection, Pertussis, and Yersinia infection.

Sources: UniProt * Open Targets * KEGG

2. Clinical Trials

Source: [ClinicalTrials.gov](https://clinicaltrials.gov)

41

Total Studies

14

Completed

12

Ongoing / Active

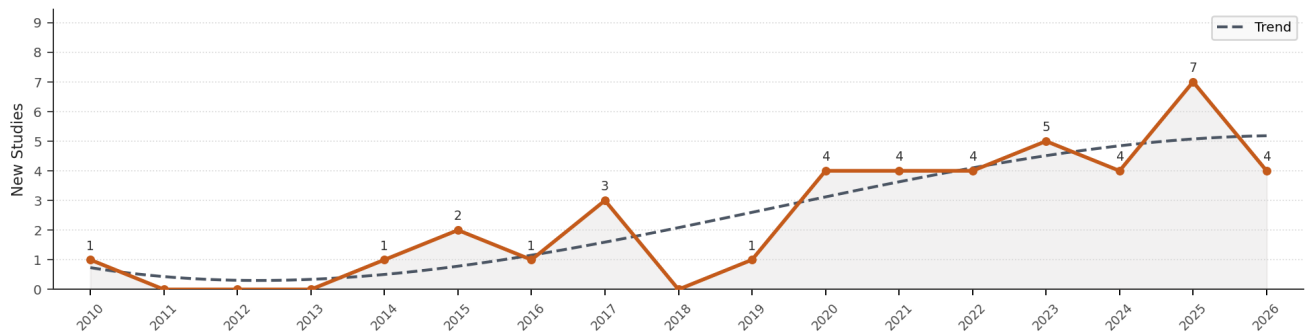
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Withdrawn

15

Other

Clinical Trials Started per Year



Clinical trials started per year with polynomial trend line.

Associated Conditions / Diseases

Condition	Studies
Atopic Dermatitis	5
Rheumatoid Arthritis	4
Hidradenitis Suppurativa	4
Healthy	3
Myelodysplastic Syndrome	3
Chronic Lymphocytic Leukemia	2
Healthy Volunteer	2
COVID-19 Pneumonia	1
Pneumococcal Infections	1
Vaccine-Preventable Diseases	1
Invasive Pneumococcal Disease, Recurrent Isolated	1

Condition	Studies
Relapsed and Refractory Acute Myeloid Leukemia (RR-AML)	1
Acute Myeloid Leukemia	1
Acute Myeloid Leukemia Arising From Previous Myelodysplastic/Myeloproliferative Neoplasm	1
Acute Myeloid Leukemia Post Cytotoxic Therapy	1

All Interventions (unique entries from trial data)

All unique intervention names recorded across clinical trials for this target, with the NCT ID(s) of the studies they appear in.

Intervention Name	NCT ID(s)
AZD2962	NCT07064122
AZD6793	NCT05662033, NCT07082738, NCT07433062
Emavusertib	NCT05685602, NCT07107750, NCT07463820
Pembrolizumab	NCT05669352
PF-06650833	NCT04933799
SAR444656 (KT-474)	NCT06028230, NCT06058156
Zabedoseritib (BAY1834845)	NCT05656911

Combination Therapies / Products

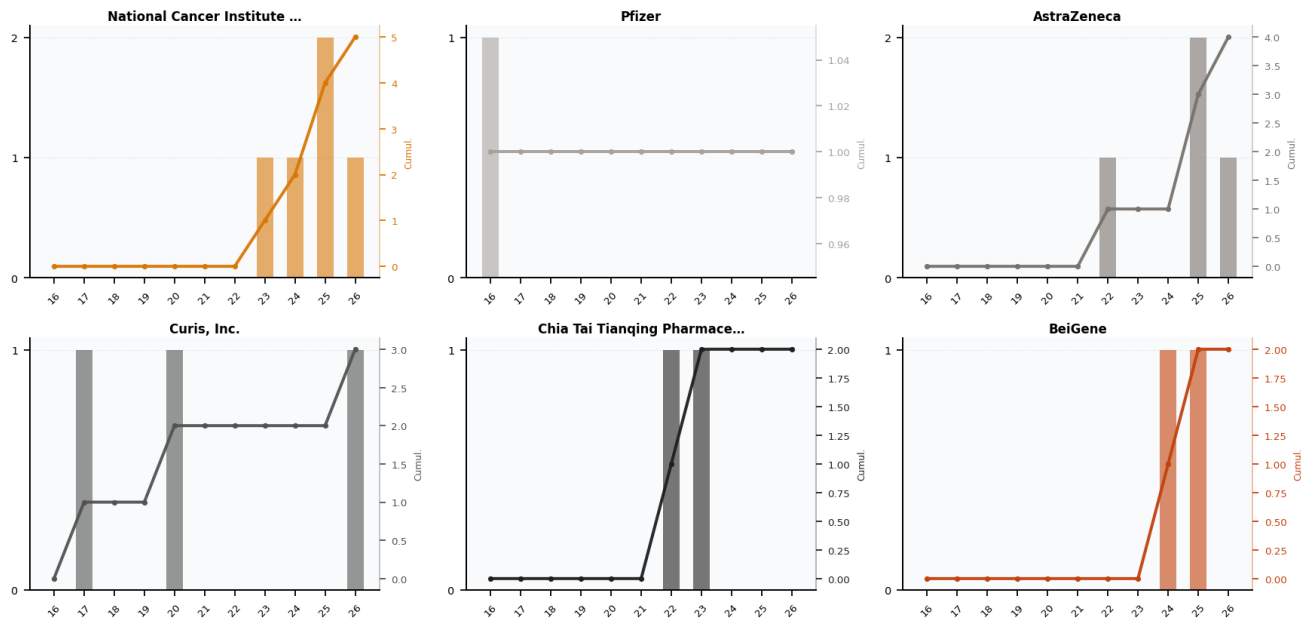
Studies where two or more interventions passed the drug-name filter. Each row represents one study.

Interventions	NCT ID
Azacitidine + Emavusertib + Enasidenib + Fludarabine + Gemtuzumab Ozogamicin + Gilteritinib + Liposome-encapsulated Daunorubicin-Cytarabine + Olutasidenib + Positron Emission Tomography + Venetoclax	NCT05564390
Emavusertib + Pembrolizumab + Positron Emission Tomography	NCT06439836
Emavusertib + Positron Emission Tomography	NCT06696768

Interventions	NCT ID
Emavusertib + Zanubrutinib	NCT07271667

Lead Sponsors - Activity & Key Studies

Lead Sponsors — Studies per Year (last 10 yrs)



Lead sponsors - annual count (bars) and cumulative total (line).

Bars = annual study count * Line = cumulative total * Last 10 years.

National Cancer Institute (NCI)

Study Title	NCT ID	Phase	Status	Start	End
A Trial Comparing Three Different Treatment Options for Adults With Low-Ris...	NCT07463820	PHASE2	NOT_YET_RECRUITING	2026-06-19	2027-12-31
Clinical Trial of an Anti-cancer Drug, CA-4948 (Emavusertib), in Combinatio...	NCT06696768	PHASE1	RECRUITING	2025-06-23	2026-12-31

Study Title	NCT ID	Phase	Status	Start	End
Pembrolizumab Plus CA-4948 for the Treatment of Patients With Progressive M...	NCT06439836	PHASE1	RECRUITING	2025-05-05	2027-06-01
MYELOMATCH: A Screening Study to Assign People With Myeloid Cancer to a Tre...	NCT05564390	PHASE2	RECRUITING	2024-06-18	2029-05-15
CA-4948 Added to Standard Chemotherapy to Treat Metastatic or Unresectable ...	NCT05685602	PHASE1	SUSPENDED	2023-06-12	2027-04-30

Pfizer

Study Title	NCT ID	Phase	Status	Start	End
Safety and Efficacy of Pf-06650833 In Subjects With Rheumatoid Arthritis, W...	NCT02996500	PHASE2	COMPLETED	2016-11-10	2018-08-15

AstraZeneca

Study Title	NCT ID	Phase	Status	Start	End
A Study to Evaluate the Effect of AZD6793 on the Pharmacokinetics and Pharm...	NCT07433062	PHASE1	SUSPENDED	2026-01-14	2026-06-18
A Study of AZD2962, an IRAK4 Inhibitor (IRAK4 [a Body Protein] Blocker), in...	NCT07064122	PHASE1	RECRUITING	2025-11-03	2028-10-20

Study Title	NCT ID	Phase	Status	Start	End
A Study to Investigate the Effect of AZD6793 in Participants With Moderate ...	NCT07082738	PHASE2	RECRUITING	2025-06-27	2027-11-05
A Study to Investigate the Safety, Tolerability, and Pharmacokinetics (PK) ...	NCT05662033	PHASE1	COMPLETED	2022-12-05	2024-10-29

Curis, Inc.

Study Title	NCT ID	Phase	Status	Start	End
A Study of Emavusertib + An Approved Bruton Tyrosine Kinase Inhibitor (BTKi...	NCT07271667	PHASE2	RECRUITING	2026-04-23	2027-11
Dose Escalation/ Expansion Study of CA-4948 as Monotherapy in Patients With...	NCT04278768	PHASE1 / PHASE2	SUSPENDED	2020-07-06	2026-04-01
CA-4948-101: Open-Label, Dose Escalation and Expansion Trial of Emavusertib...	NCT03328078	PHASE1 / PHASE2	RECRUITING	2017-12-28	2026-10

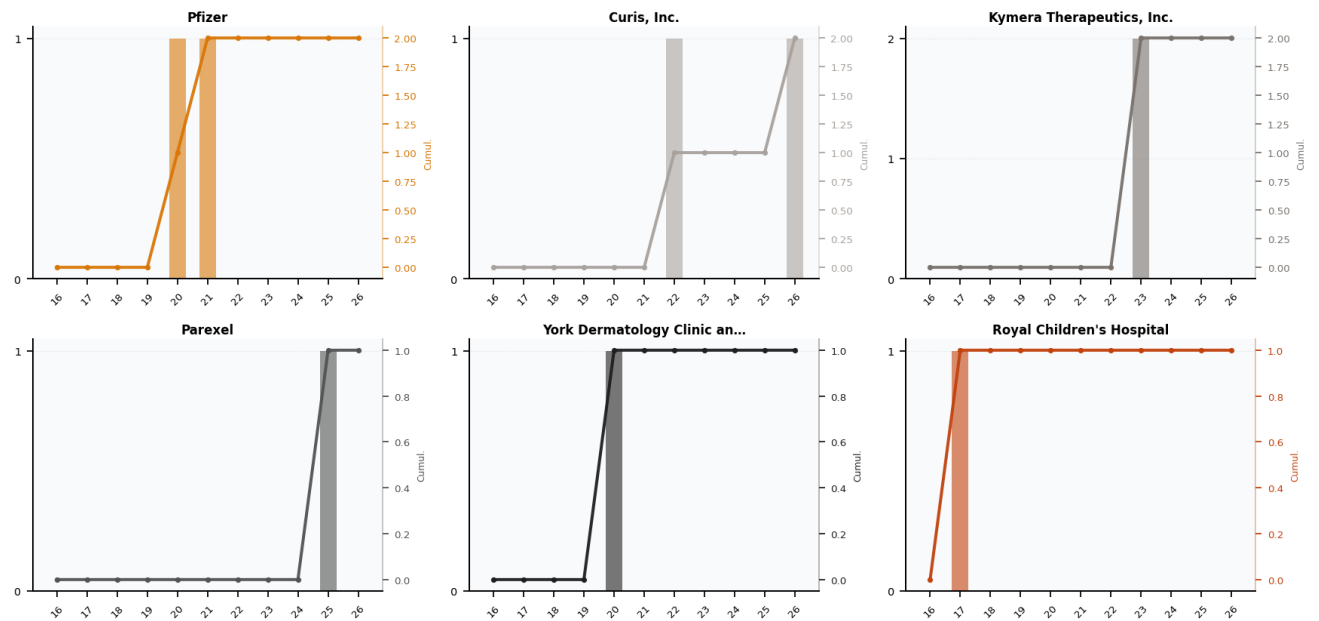
Chia Tai Tianqing Pharmaceutical Group Co., Ltd.

Study Title	NCT ID	Phase	Status	Start	End
A Clinical Study of TQH3821 Tablets in the Treatment of Treated Rheumatoid ...	NCT05849727	PHASE2	TERMINATED	2023-08-30	2025-01-15
A Clinical Trial of Safety and Tolerance of TQH3821 Tablets in Adult Health...	NCT05380934	PHASE1	COMPLETED	2022-05-29	2023-05-20

Study Title	NCT ID	Phase	Status	Start	End
A Study Investigating the Efficacy and Safety of BGB-45035 Versus Placebo i...	NCT07100938	PHASE2	TERMINATED	2025-08-21	2026-05-12
A Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Pharmac...	NCT06342713	PHASE1	RECRUITING	2024-06-20	2026-05-23

Collaborating Institutions - Activity & Key Studies

Collaborating Institutions — Studies per Year (last 10 yrs)



Collaborating institutions - annual count and cumulative total.

Bars = annual study count * Line = cumulative total * Last 10 years.

Pfizer

Study Title	NCT ID	Phase	Status	Start	End
IRAK 4 Inhibitor (PF-06650833) in Hospitalized Patients With COVID-19 Pneum...	NCT04933799	PHASE2	UNKNOWN	2021-01-06	2022-05-06
IRAK4 Inhibition in Treatment of COVID-19 With ARDS (I-RAMIC)	NCT04575610	PHASE2	TERMINATED	2020-11-27	2021-10-06

Curis, Inc.

Study Title	NCT ID	Phase	Status	Start	End
Emavusertib (CA-4948) in Combination With Cisplatin, Gemcitabine, and Durva...	NCT07107750	PHASE1	RECRUITING	2026-03-27	2032-05-31
Treatment of Anemia in Patients With Very Low, Low or Intermediate Risk Mye...	NCT05178342	PHASE2	TERMINATED	2022-01-01	2024-07-31

Kymera Therapeutics, Inc.

Study Title	NCT ID	Phase	Status	Start	End
Phase 2 Study of SAR444656 in Adult Participants With Moderate to Severe At...	NCT06058156	PHASE2	TERMINATED	2023-11-09	2025-11-20
A Phase 2 Study to Evaluate the Efficacy and Safety of SAR444656 Compared W...	NCT06028230	PHASE2	TERMINATED	2023-09-29	2025-11-12

Parexel

Study Title	NCT ID	Phase	Status	Start	End
A Study of AZD2962, an IRAK4 Inhibitor (IRAK4 [a Body Protein] Blocker), in...	NCT07064122	PHASE1	RECRUITING	2025-11-03	2028-10-20

York Dermatology Clinic and Research Centre

Study Title	NCT ID	Phase	Status	Start	End
Evaluation of Cutaneous and Circulating Inflammatory Biomarkers in Hidraden...	NCT04440410	N/A	COMPLETED	2020-05-28	2021-03-24

Royal Children's Hospital

Study Title	NCT ID	Phase	Status	Start	End
What is the Incidence of an Immune Disorder in Children With Invasive Pneum...	NCT03815357	N/A	COMPLETED	2017-01-31	2021-06-30

Withdrawn Studies & Reasons

No withdrawn studies found.

Trial & Patent Data Files

Trials CSV (21 records)

IRAK4/csvs/IRAK4_trials.csv

Patents CSV (0 records)

IRAK4/csvs/IRAK4_patents.csv

Sources: *ClinicalTrials.gov (trials) * EPO Espacenet (patents)*

3. Drug Discovery Landscape

4

Active Dimensions

6

Emerging Dimensions

10

Total Shown

For a given protein target, this analysis identifies drug-discovery dimensions explored in the literature and evaluates their volume and recency to highlight factors most relevant to the program. It focuses on publications from the past 5 years across key drug-discovery dimensions.

Landscape Summary

Property	Value
Active Dimensions	4
Emerging Dimensions	6
Total Shown	10

Dimensions Detail

Rank	Area	Dimension	Status	PMIDs
1	Modality	Degraders / PROTACS / Molecular Glues	Active	31620217 38049464 39151120 37197458 41847655
2	Combination Therapy	Combination / Synergy	Active	41863454 40536312 38666914 34248990 28750570
3	Hit Discovery & Optimisation	SAR / Lead Optimisation	Active	39515709 34427398 30818109 20429783
4	Mutations & Variants	Activating / Driver Mutations	Emerging	32532880 29468521 21179087 12885415
5	Preclinical & Clinical	Clinical Candidate / Trial	Active	34424598 34423919
6	Binding Mode	Type II / DFG-out / Inactive State	Emerging	37172355
7	Selectivity Strategy	Dual / Multi-Target	Emerging	36540413
8	Hit Discovery & Optimisation	Fragment-Based / HTS	Emerging	38913701

Rank	Area	Dimension	Status	PMIDs
9	Binding Site	Allosteric Site / Modulator	Emerging	33238146
10	Selectivity Strategy	Selectivity Profiling / Off-Target Panel	Emerging	30867219

4. ChEMBL

This section curates approved and investigational small-molecule drugs associated with the target protein from ChEMBL and maps them to corresponding PDB structures. Each molecule is annotated with its highest reported binding activity against the primary target and its selectivity profile relative to off-target proteins identified in ChEMBL bioactivity data.

Molecules

Name	Phase	Best pChEMBL	Target PDB	Selectivity (delta, top off-target)	Off-Targets (top 4)
ZIMLOVISERTIB	Phase 2	10.00 [Ref]	human: 5UIU, 8WTF	No off-target data in ChEMBL	-
EDECESERTIB	Phase 2	9.00 [Ref]	-	No off-target data in ChEMBL	-
EMAVUSERTIB	Phase 2	7.50 [Ref]	human: 7C2V	No off-target data in ChEMBL	-
KT-474	Phase 1	7.39 [Ref]	-	No off-target data in ChEMBL	-
LESTAURTINIB	Phase 3	8.77 [Ref]	-	delta=-0.64 (PHKG1 pC=9.41) [Ref]	PHKG1, MAP3K19, FLT3, JAK2
QUIZARTINIB	Phase 4	8.69 [Ref]	-	delta=-3.01 (FLT3 pC=11.70) [Ref]	FLT3
PONATINIB	Phase 4	7.17 [Ref]	human: 6EG9	delta=-3.23 (FLT3 pC=10.40) [Ref]	FLT3, ABL1, HCK

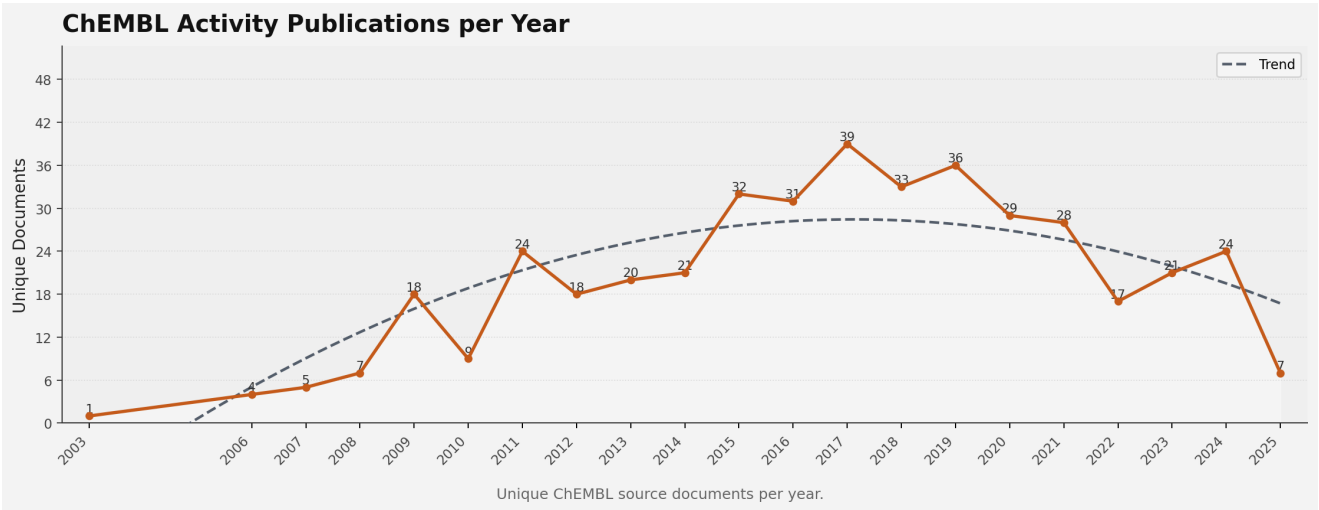
* Name: preferred drug name. * Phase: highest clinical phase reached (1-4; 4 = approved). * Best pChEMBL: highest -log10(molar) activity value in ChEMBL against the primary target; [\[Ref\]](#) links to the source document. * Target PDB: PDB entry IDs where the ligand is co-crystallised with the primary target (human / mouse). * Selectivity: difference between on-target pChEMBL and the best off-target pChEMBL (positive = more potent on-target). Top off-target gene and its activity shown in parentheses; [\[Ref\]](#) links to the source document. * Off-Targets: up to 4 off-target gene symbols ranked by pChEMBL whose bioactivity exceeded the potency threshold in ChEMBL.

Top PDB Structures for IRAK4 (Resolution < 3.0 Å)

The following PDB structures are available for IRAK4 with resolution below 3.0 Å, suitable for structure-based drug design and docking studies. Resolution is reported in Angstroms; lower values indicate higher-quality electron density maps.

PDB ID	Resolution (Å)
2NRU	2.0
2OIB	2.0
2NRY	2.15
2O8Y	2.4
2OIC	2.4

ChEMBL Activity Data Processing



IRAK4 - ChEMBL studies published per year (pre-threshold).

This section summarizes molecular activity data reported in ChEMBL across multiple activity metrics. It highlights publication trends over time for the target, quantifies the number of studies/documents associated with each metric, and includes scaffold-level information for molecules reported in each document within the accompanying CSV files.

15,015

Molecules in

10,386

Molecules after filter

303

Studies passing

Activity Thresholds & Record Counts

Standard Type	Threshold	Docs (passed)	Mols in docs
Activity	>= 50	76	130

Standard Type	Threshold	Docs (passed)	Mols in docs
EC50	<= 10,000 nM	4	155
ED50	<= 10,000 nM	1	2
IC50	<= 10,000 nM	170	8,472
Inhibition	>= 50	62	137
Kd	<= 1,000 nM	9	275
Ki	<= 1,000 nM	16	1,476

Activity thresholds are applied to retain only studies whose most potent molecule meets a minimum activity criterion - studies where even the best molecule fails to reach this bar are excluded from downstream analysis. 'Mols in docs' counts all unique molecules across surviving documents for that standard type, not just the single best molecule selected per document.

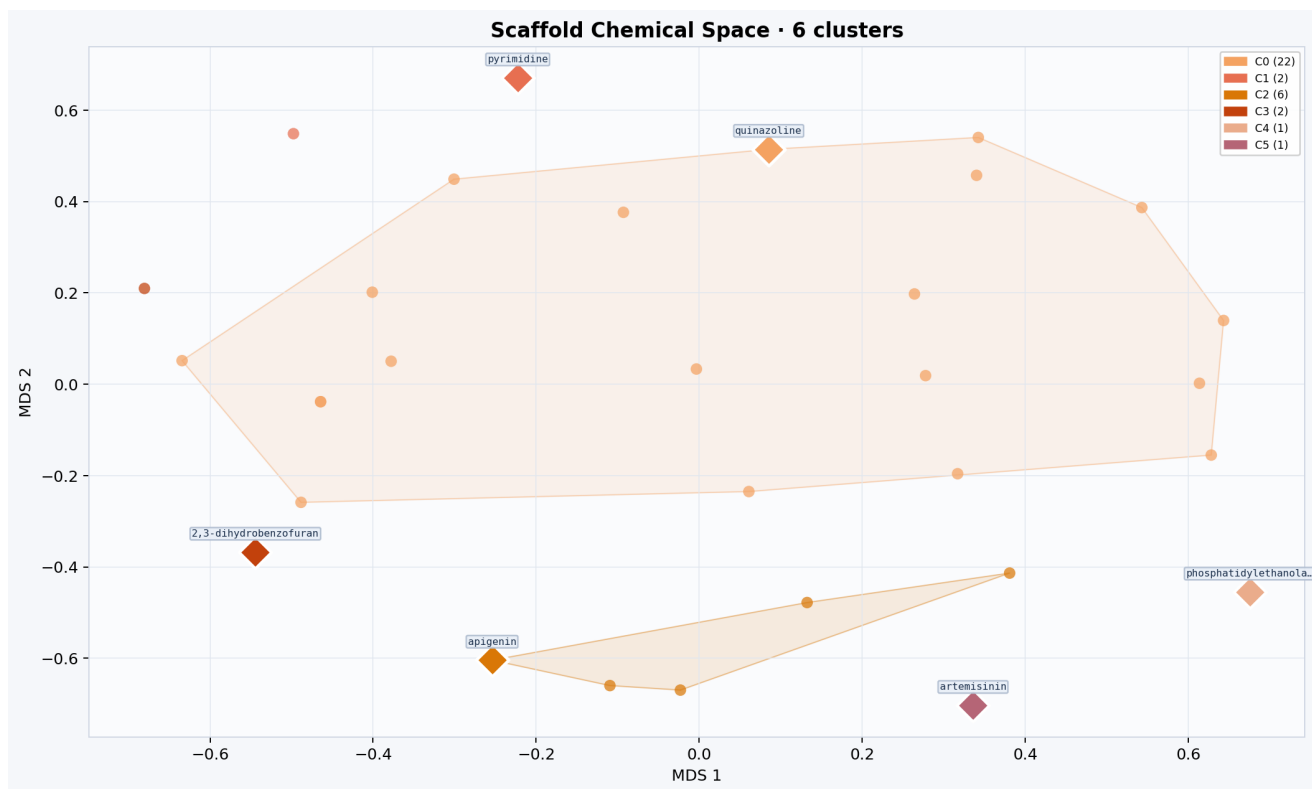
5. Scaffold Study

Scaffold mining uses PubMed literature for the target, screens titles with a local LLM, extracts scaffold names, fetches SMILES from PubChem and CACTUS, then deduplicates by name. The sorted CSV is the primary input for clustering and scaffold hopping.

Scaffolds by Publication Year

Year	Scaffolds
2026	3H-spiro[benzofuran-2,4'-piperidine]
2025	pyrrolo[2,3-d]pyrimidine, apigenin
2024	pyridone, imidazo[1,2-a]pyrimidine, artemisinin, 2,4'-dihydroxybenzophenone
2023	1,3-oxazole, 2,3-dihydrobenzofuran, dihydrofuro[2,3-b]pyridine
2022	pyrimidopyridone, pyrimidine
2021	indazolamine, chalcone, fisetin
2020	imidazo[1,2-b]pyridazine, quinazoline
2019	quinazoline, pyrazolopyrimidine
2018	pyrrolotriazine
2017	apigenin, quinazoline, pyrrolopyrimidine, 4,6-diaminonicotinamide, isoquinoline
2016	phosphatidylethanolamine
2015	1H-pyrazol-4-yl carboxamide, 2,6-diaminopyrimidin-4-one, diaminopyrimidine, pyrazolo[1,5-a]pyrimidine, diindolylmethane
2014	indolo[2,3-c]quinoline
2013	quercetin
2008	imidazo[1,2-a]pyridine

Scaffold Clustering



Scaffold chemical space coloured by Butina cluster. Diamonds = medoids.

Clustering Parameters

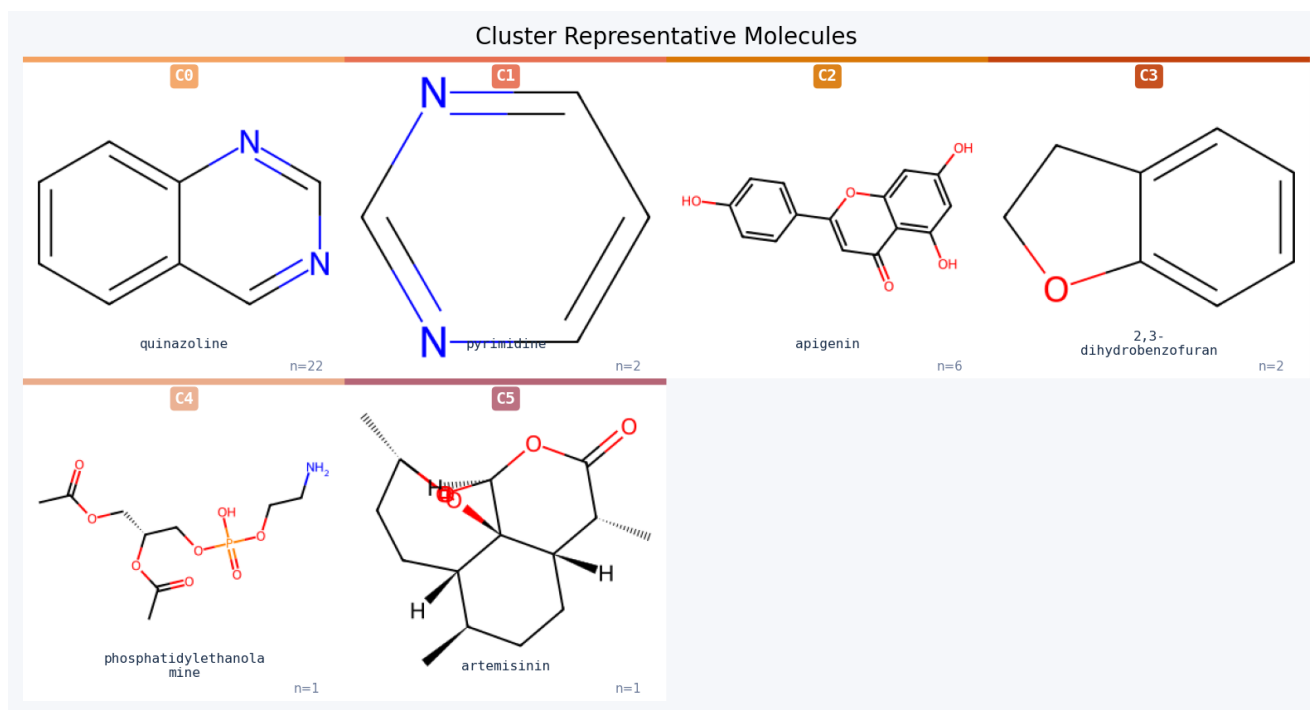
Property	Value
Dimensionality Reduction	MDS
Similarity Method	Butina / Tanimoto
Fingerprint	Morgan r=2, 2048 bits
Similarity Cutoff	0.15
Clusters Found	6
Valid Molecules	34

Selected Central Scaffold

Property	Value
Scaffold Name	quinazoline
SMILES	<chem>C1=CC=C2C(=C1)C=NC=N2</chem>
Cluster summary CSV	0 records - IRAK4/csvs/IRAK4_scaffolds_sorted_cluster_summary.csv

The cluster with the highest number of unique scaffolds was selected as the primary cluster. Its medoid - the molecule with the highest average Tanimoto similarity to all other cluster members - was chosen as the central

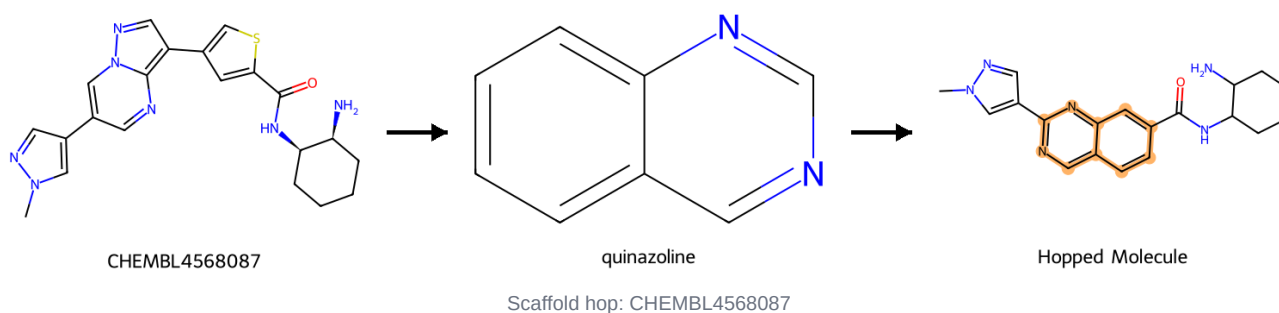
scaffold for downstream scaffold hopping. The chemical space map shows all scaffolds coloured by cluster with convex hulls; the cluster centres panel shows the representative structure per cluster.

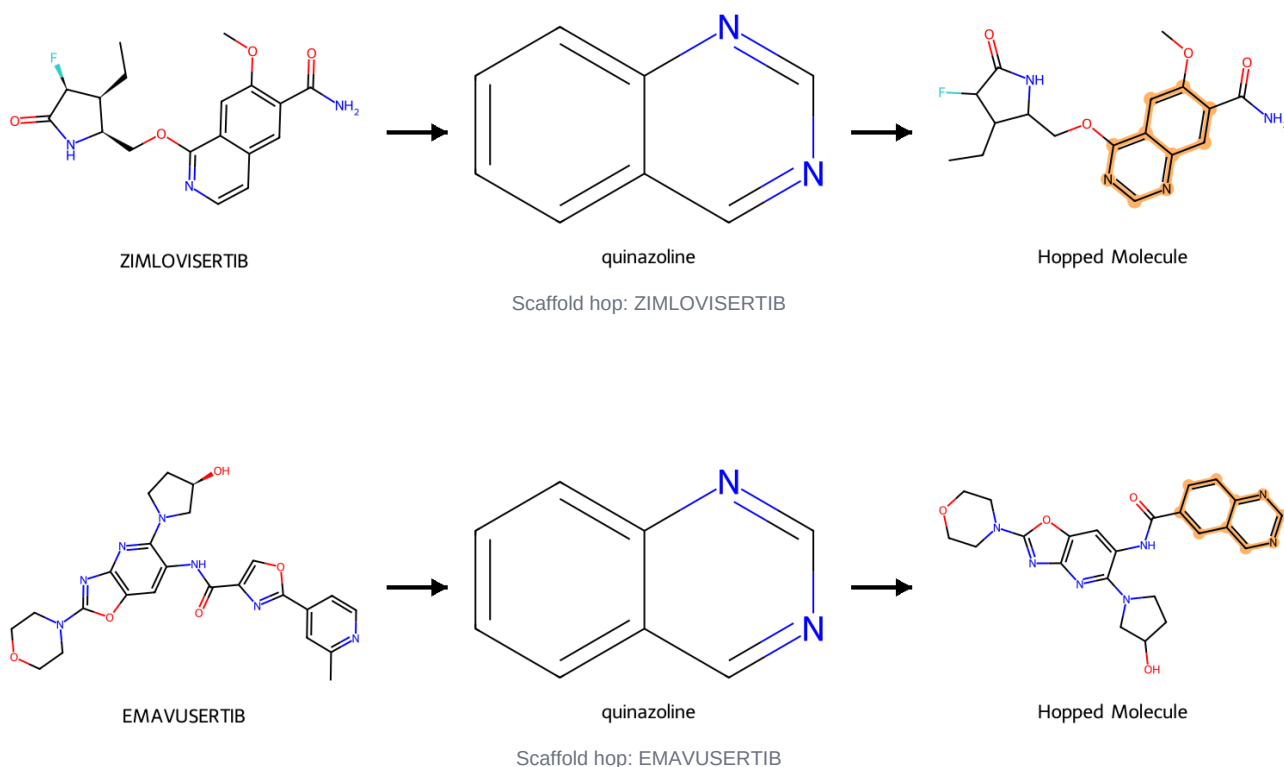


Cluster representative scaffolds (medoids).

Scaffold Hopping & Fragment Replacement

The central scaffold identified from the clustering analysis was selected for scaffold hopping. Molecules from the ChEMBL IC50 dataset exhibiting the highest scaffold pharmacophore similarity to the central scaffold, along with selected reference molecules from the approved/investigational compounds list, were used as the basis for the scaffold hopping process.





Fragment Replacement (FARE)

Fragments with physicochemical properties similar to those already present were extracted from the fragment library and attached to the scaffold using the selected molecule as the reference framework.

Generated Molecule Files

CSV File	Path	Molecules Generated
quinazoline_CHEMBL4568087_FARE_generated_molecules.csv	IRAK4/generated\quinazoline_CHEMBL4568087_FARE_generated_molecules.csv	2068
quinazoline_EMAVUSERTIB_FARE_generated_molecules.csv	IRAK4/generated\quinazoline_EMAVUSERTIB_FARE_generated_molecules.csv	1456
quinazoline_ZIMLOVISERTIB_FARE_generated_molecules.csv	IRAK4/generated\quinazoline_ZIMLOVISERTIB_FARE_generated_molecules.csv	5472

Generated Data Files

The following data files were generated during this analysis and are archived for reference and further processing.

File Name	Type	Records	Location
IRAK4_trials.csv	CSV	varies	IRAK4
IRAK4_patents.csv	CSV	varies	IRAK4
IRAK4_scaffolds_sorted.csv	CSV	varies	IRAK4
IRAK4_scaffolds_sorted_cluster_summary.csv	CSV	varies	IRAK4